

various cloud resources, including virtual machines, databases, storage, and applications. Figure 1 provides a brief overview of a VPC.

Critical differences between a VPC and a private cloud

VPC is a virtual network within a public cloud, offering scalability, flexibility, and cost efficiency while still providing strong security and isolation measures. It relies on the public cloud provider's infrastructure, making it a good choice for organisations looking for a balance between cost, scalability, and security. A private cloud is a dedicated environment exclusively for a single organisation, offering maximum control, customisation, and security. Table 1 lists the critical differences between a VPC and a private cloud.

Due to their flexibility, security, and scalability, VPCs have several use cases across different industries and applications (Figure 2).

Deploying VPC and web server services on AWS cloud

Establishing a VPC and running a web server on AWS cloud requires providing a secure and separate network

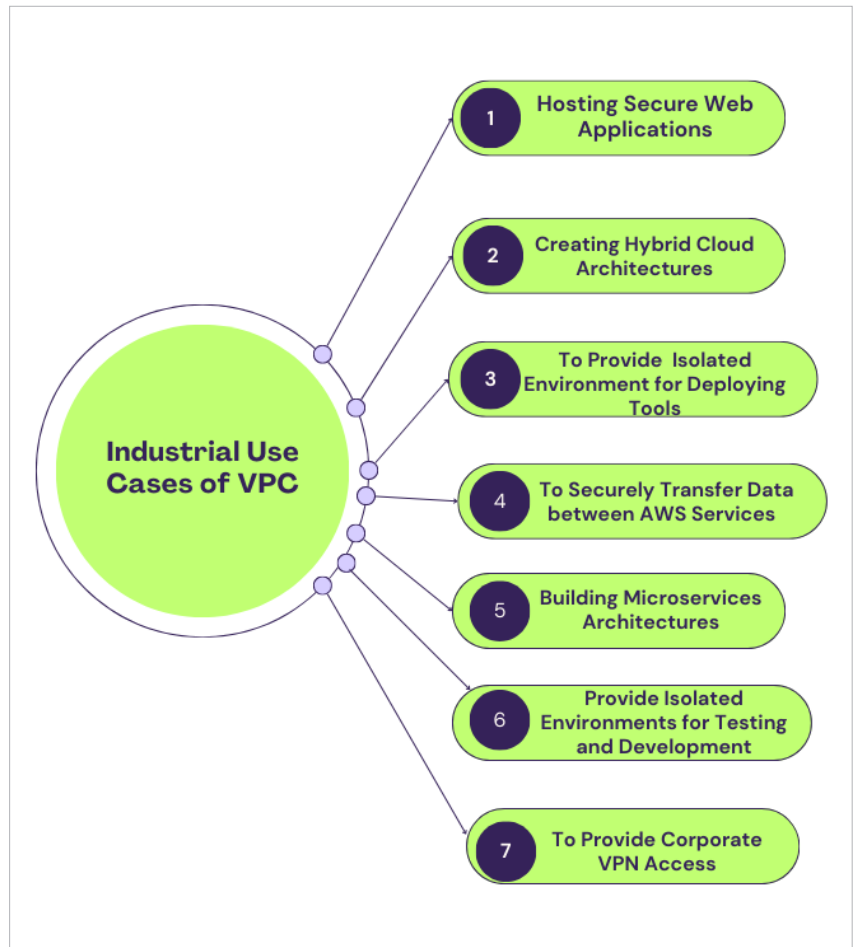


Figure 2: Industrial use cases of VPCs

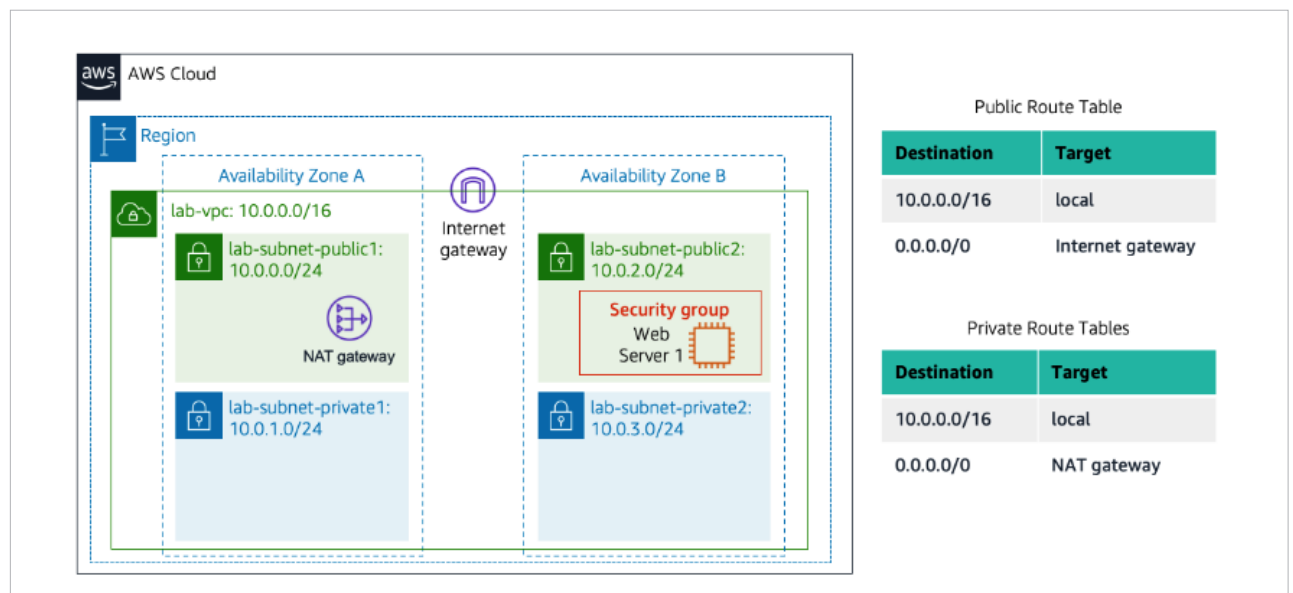


Figure 3: Experimental architectural diagram of VPC